

Expectation and Outcome: A Tracking-Based Framework for Evaluating Stolen Bases

Team 305

Abstract

Stolen base analysis has traditionally relied on scorekeeper-recorded outcomes, which record whether a runner advanced but not the conditions that produced the result. Genuine attempts are indistinguishable from opportunistic advances on wild pitches or passed balls, and the athleticism on both sides of the play goes unmeasured. A common shortcut compounds the problem: requiring a catcher throw to identify an attempt discards uncontested steals, biasing the sample toward contested plays.

This project uses SMT Data Challenge tracking data to identify first-to-second steal attempts directly from movement rather than recorded outcomes. Attempts are detected through base-state filtering, runner closing speed relative to pitch release, and progress toward second base, with thresholds validated against observed displacement distributions. From the same tracking stream, we derive runner jump and sprint speed alongside catcher pop time, exchange time, and throw velocity. These metrics feed live top-10 scoreboards for runner and catcher performance, while a safe/out prediction model estimates each runner's Safe% and flags plays that beat expectation.

The result is an unbiased attempt population with continuous measurements of both runner and defender, letting clubs credit the execution behind each steal rather than only its outcome.

Introduction

Baseball broadcasts and scoreboards have increasingly used player-tracking metrics to explain aspects of the game in real time. For the running game, MLB's Statcast already provides fan-facing metrics such as [sprint speed](#), which measures a player's fastest one-second running window in feet per second, and [catcher pop time](#), which measures the time from the pitch reaching the catcher to the throw reaching the glove at the base. Statcast's baserunning leaderboards also go beyond simple stolen base totals by incorporating measurements such as runner speed, distance, and success probability. These tools show that the running game can be evaluated as a sequence of measurable events rather than only as a final "safe" or "out" result.

Many of these metrics, however, are presented as season-long leaderboards, broadcast graphics, or post-play summaries. In a Minor League scoreboard setting, the opportunity is to turn each steal attempt into an immediate, understandable moment for fans in the stadium. A fan may see a

runner steal second base without knowing whether the play was impressive because of the runner's speed, jump, the catcher's exchange, throw velocity, slide, or tag.

This project builds a live scoreboard framework that recognizes the top 10 steal-related events across four metrics: runner top speed, runner jump, catcher pop time, and catcher throw velocity. It also uses a safe/out prediction model to flag plays that beat expectations, letting the scoreboard highlight moments such as a great slide or strong tag.

Data Preprocessing

Filtering process

Stolen base attempts were identified from the base-state recorded in the lineups table, restricting candidates to plays with a runner on first and second base unoccupied so that any advance could be attributed unambiguously to that runner; plays involving a batted ball or pickoff throw were excluded, as advances in those cases reflect batted-ball or pickoff action rather than an independent decision to steal. In addition, forced advances on walks or hit-by-pitch were explicitly removed. Confirmation then relied on tracking data rather than box-score inference: displacement of the runner was measured as progress toward second base with a threshold of approximately forty-five feet, and the runner's moment of commitment. The presence of a catcher throw was deliberately not used as a detection criterion, since requiring a throw would discard uncontested steals and bias the labels toward contested outcomes.

Metrics

Runner top speed was defined as the fastest short, sustained burst of speed observed during the steal attempt. Runner speed was calculated from changes in field position over time, and top speed was reported as the highest one-second rolling average of runner speed. Using this rolling average helped reduce the effect of frame-level noise while still capturing the runner's peak speed during the play.

Runner jump was defined as the amount of speed gained in the first 0.4 seconds after pitch release. It was calculated as the runner's speed closest to 0.4 seconds after pitch release minus the runner's speed closest to pitch release, reported in feet per second gained. A higher value represents a more explosive early acceleration.

Catcher pop time was defined for each confirmed attempt as the interval between the catcher's acquisition of the pitch and the middle infielder's receipt of the throw at second base, drawn

directly from event timestamps. Restricting the last event to acquisition by the second baseman or shortstop ensures the measurement reflects the full transfer-and-delivery sequence relevant to the runner's outcome, and excludes plays in which no throw was made or the ball was received elsewhere.

Catcher throw velocity was computed from frame-by-frame ball position over the interval between catcher release and infielder acquisition, taking successive three-dimensional displacements divided by elapsed time and converting to miles per hour. Both peak and mean velocity were kept: peak velocity reflects the catcher's arm strength, while mean velocity reflects how quickly the ball actually covered the distance to second base.

Limitations

This project has several limitations related to the data sample size, metric construction, and modeling approach. First, runner and catcher metrics are estimated from tracking data that can contain frame-level noise, which is why runner top speed uses a short rolling average rather than a single-frame maximum.

Similarly, not every steal attempt contained clean tracking information. Some plays had missing frames, unclear timing events, or unusual game situations, so certain attempts had to be filtered out or excluded from leaderboards. We had to balance keeping as many plays as possible against using only plays where metrics could be reliably calculated.

Methodology

Leaderboard construction

The first part of the project converts raw tracking data into live scoreboard leaderboards built from the four event-level metrics defined above, so the scoreboard can surface the specific parts of a play that made it impressive rather than only its safe/out result.

Plays were ranked independently within each category, and a steal attempt qualified for the scoreboard if it ranked inside the top 10 of any category. If a play qualified in multiple categories, the system prioritized the strongest trigger by rank and displayed the corresponding leaderboard.

Prediction model

The second part of the analysis used the calculated runner and catcher metrics to model whether a steal attempt should end safe or out. The model's purpose was not only to predict the result of a steal attempt, but also to create a baseline expectation for each play. When the actual result differed from the model's expectation, that disagreement became a possible fan-facing explanation moment.

Before selecting the final model, logistic regression, a decision tree, and a random forest were compared on a held-out test set. The models were evaluated using accuracy, balanced accuracy, and test AUC. Although the decision tree produced the highest test AUC, logistic regression was selected as the final model because it produced smoother and more interpretable safe probability estimates. This was more useful for a scoreboard setting, where the goal was to display a clear Safe% rather than only classify the play as safe or out.

So, the final model was a logistic regression model predicting the probability that the runner would be safe. The response variable was a binary version of the steal outcome, where safe plays were coded as 1 and caught stealing plays were coded as 0. The model used four tracking-based inputs: runner top speed, runner jump, catcher pop time, and catcher throw velocity. Runner top speed and jump describe the offensive side of the steal attempt, while pop time and throw velocity describe the defensive response from the catcher.

After the final logistic regression model was fit, each steal attempt received a predicted Safe%. Plays with a high predicted Safe% that ended in an out were labeled as possible **“GOOD TAG!”** moments, while plays with a low predicted Safe% that ended safe were labeled as possible **“GOOD SLIDE!”** moments. These labels are not meant to definitively judge the slide or tag, but instead act as simple scoreboard prompts that tell fans when the play beat the model's expectation and may deserve extra attention.

Because the model is imperfect, its mistakes are built into the design rather than treated only as failures. If the model gives a runner a high probability of being safe but the runner is called out, the system flags a possible "good tag" or strong defensive execution; if it gives a low probability but the runner beats the throw, it flags a possible "good slide" or runner-created steal. Used this way, the model does not need to be correct every time to be useful: the leaderboards explain what was physically impressive, and the model adds context for when an outcome was surprising given the available tracking information.

Assumptions

This approach assumes that the calculated tracking metrics are reliable enough to compare across steal attempts after filtering and smoothing. Runner top speed, runner jump, catcher pop time, and catcher throw velocity are treated as meaningful summaries of each play, even though the raw tracking data can contain frame-level noise or missing values.

For the prediction model, we assume that the included runner and catcher metrics capture part of the relationship between player movement and steal outcome. The model does not assume that these four variables explain every safe or out result. Other factors, such as pitcher time to the plate, runner lead distance, throw accuracy, pitch location, tag placement, slide quality, and umpire judgment, are not fully captured in the model.

Results

In the final workflow, each steal attempt is checked against the season-long top 10 for the four metrics; when a play qualifies, the scoreboard renders the leaderboard for that category. Figures 1 and 2 show the same framework triggered by two different metrics: runner top speed and catcher pop time.

TOP RUNNER SPEEDS			
Fastest sustained bursts during steal attempts			
RANK	PLAYER	GAME	VALUE
#1	Player ALC-9240	D145: ALC-ARN	21.15 mph
#2	Player PWH-0771	D211: PWH-ARN	20.85 mph
#3	Player ANI-7878	D166: SZJ-ANI	20.74 mph
#4	Player JPB-0696	D113: JPB-ANI	20.49 mph
#5	Player EXB-2802	D137: EXB-ARN	20.46 mph
#6	Player PHD-0812	D124: VAS-PHD	20.45 mph
#7	Player MEX-6247	D125: MEX-ANI	20.39 mph
#8	Player JJP-9950	D082: JJP-PHD	20.17 mph
#9	Player MEX-9074	D127: MEX-ANI	20.13 mph
#10	Player VAS-1487	D199: JGO-VAS	20.12 mph

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Figure 1. Runner Top Speed Leaderboard. An example of the scoreboard graphic as it appears when a steal attempt ranks among the season's fastest runners. The board lists the top 10 attempts by runner top speed with the highest one-second rolling average of runner speed recorded during a steal, sorted fastest to slowest. Each row shows rank, the runner's player ID, and the speed value, and the attempt that just occurred is highlighted in gold: here, runner ANI-7878 at 20.74 mph, entering at #3. A play triggers this board whenever it cracks the category's top 10, so the graphic populates directly from the live tracking feed as new attempts happen. Because top speed is smoothed over one second, momentary tracking spikes do not inflate the ranking.

FASTEST POP TIMES			
Quickest catcher throws to second base			
RANK	PLAYER	GAME	VALUE
#1	PHD-1960	D148: VHA-PHD	1.78 sec
#2	JPB-2826	D115: JPB-ANI	1.85 sec
#3	XGM-5318	D111: XGM-PHD	1.85 sec
#4	PHD-0812	D124: VAS-PHD	1.85 sec
#5	VAS-7569	D133: ARN-VAS	1.85 sec
#6	EXB-6199	D137: EXB-ARN	1.85 sec
#7	PHD-4573	D083: RYU-PHD	1.9 sec
#8	OER-0542	D105.5: OER-ANI	1.9 sec
#9	MSE-2365	D122: MSE-ANI	1.9 sec
#10	PHD-0812	D124: VAS-PHD	1.9 sec

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Figure 2. Catcher pop time leaderboard. The same framework triggered by a defensive metric rather than a runner metric. Pop time is the interval from the catcher's acquisition of the pitch to the middle infielder's receipt of the throw at second base, reported in seconds, so lower values rank higher. In addition to rank, player ID, and value, this version adds a game column identifying the game the throw came from, giving fans a specific matchup to tie the play to. The triggering attempt is highlighted in gold, catcher XGM-5318 at 1.85 seconds, ranked #3. Only the title, subtitle, sorting direction, units, and the optional game column differ from Figure 1; the surrounding structure is unchanged.

The prediction model adds a second layer: after each attempt it estimates the runner's probability of being safe and compares it against the actual outcome, flagging plays that run counter to expectation as possible explanation moments. For instance, a runner given a high probability of being safe but called out can be labeled a "good tag," suggesting something outside the model, such as tag placement or throw accuracy, led to an out.

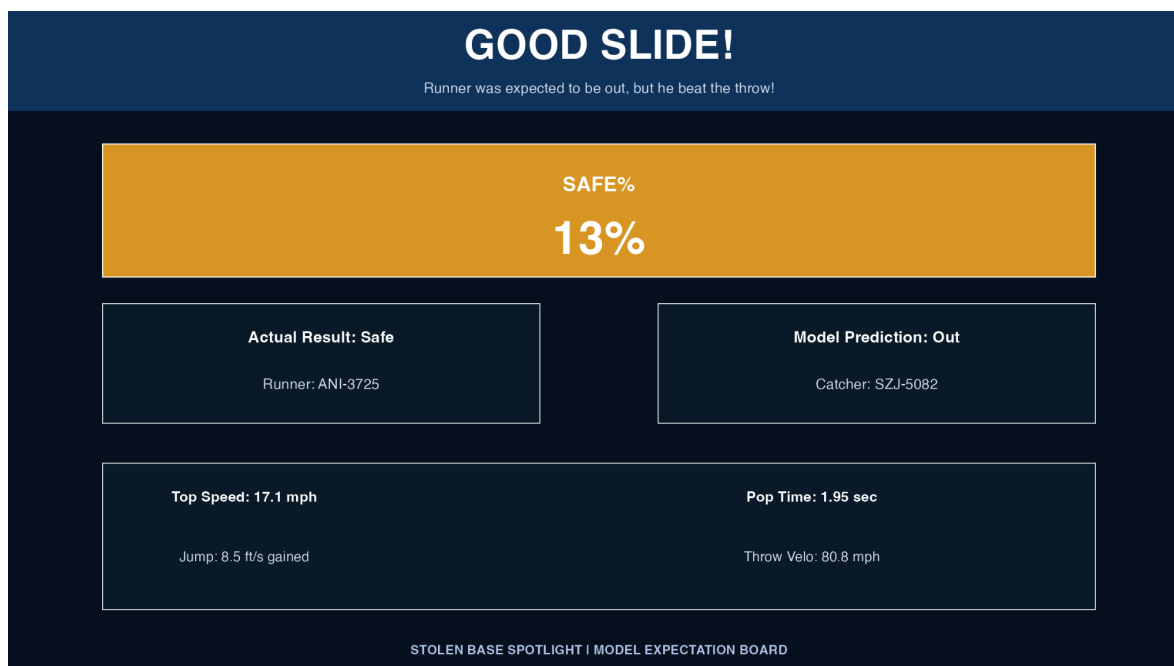


Figure 3: Model predicts high probability of out, but the runner is actually called safe: *Despite a 1.95 second pop time and an 80.8 mph throw from the catcher, the runner was called safe. Such disagreements between prediction and outcome isolate variation attributable to unmeasured factors like slide quality, tag application, and throw accuracy, and are flagged as "GOOD SLIDE!"*

Overall, the leaderboard system is the project's strongest application: the top 10 boards are easy to explain and able to highlight both offensive and defensive contributions to the running game. The prediction model works best as an explanation tool rather than a perfect safe/out classifier; by flagging plays that beat its expectation, it adds storytelling opportunities for fans. Together, they show how tracking data can be transformed into scoreboard content that is immediate, digestible, and tied to the live game environment.

Conclusion

At its core, this project takes parts of the running game that normally go unnoticed and makes them visible to fans while the play is still fresh. A stolen base is usually reduced to a single safe-or-out call, but the outcome depends on a chain of fast, hard-to-see actions: the runner's jump off first, the top speed they reach, the catcher's exchange, and the velocity of the throw to second. By measuring each of these from tracking data and surfacing the standout ones on a scoreboard, the system lets fans see why a play was impressive, not only how it ended.

This is most valuable in a Minor League setting, which typically lacks the broadcast graphics and fan-facing tracking metrics common at the major league level. The leaderboard framework compresses a large tracking dataset into a small number of immediately readable moments, and because one layout serves every metric, it can respond to whatever made a given play notable like a fast runner, an explosive jump, or a quick, strong throw. The product is scoreboard content that connects directly to the play the crowd just watched, without requiring any knowledge of the underlying data.

The prediction model extends the system from describing a play to setting an expectation for it. Among the models compared, logistic regression was selected as the final model because it produced smoother and more interpretable Safe% estimates than the more complex models. This fits the scoreboard goal better than simply maximizing classification performance. The model uses runner top speed, runner jump, catcher pop time, and catcher throw velocity to estimate the probability that the runner will be safe. When an outcome runs against that probability, the gap becomes part of the story, pointing fans toward parts of the play the tracking data does not fully capture, such as slide quality, tag placement, or throw accuracy.

Several directions remain open. The model could be refined and validated on a larger set of attempts, additional metrics could broaden the categories a play can qualify for, and the tracking-based measurements would benefit from further validation against known plays. These extensions would strengthen the system, but they do not change its central contribution: turning the invisible, split-second components of the running game into something fans can see, understand, and appreciate in real time.

Acknowledgements:

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Appendices:

Metric definitions. All spatial quantities are computed in the coordinate system defined in the SMT Data Challenge Data Description: $x = 0$ lies along the line from home plate to second base with $x > 0$ toward first base, $y = 0$ at the back of home plate with $y > 0$ toward second base, both in feet. Second base is located at $(0, 90\sqrt{2})$ and first base at $(90/\sqrt{2}, 90/\sqrt{2})$. Timestamps are in milliseconds and begin three seconds before ball release for each `play_per_game`. Runner tracking uses `player_id` 11 (runner on first base) from the player-positioning table; catcher events use `player_id` 2.

Pitch release. The timestamp of `ball_eventcode` 1 (pitch) for the play. All runner and defensive timing metrics are measured relative to this reference point.

Break time. The elapsed time, in seconds, from pitch release to the first tracking frame at which the runner's closing speed toward second base reaches 12 ft/s. This isolates the initiation of the advance from the smaller velocities produced by secondary leads.

Jump. The runner's early acceleration after pitch release, calculated as the runner's speed approximately 0.4 seconds after pitch release minus the runner's speed at pitch release. It is reported as speed gained, in mph.

Top speed. The runner's highest one-second rolling average speed during the steal attempt, calculated from frame-to-frame displacement and converted to miles per hour.

Minimum distance to second. The smallest Euclidean distance, in feet, between the runner and second base over the course of the play.

Displacement toward second. The runner's net progress along the first-to-second base path, measured as the reduction in distance to second base from the position held at pitch release.

Pop time. The elapsed time, in seconds, from the catcher's receipt of the pitch (`ball_eventcode` 2 with `player_id` 2) to receipt of the throw by the covering middle infielder (`ball_eventcode` 2 with `player_id` 4 or 6). Pop time is decomposed into two components:

- **Exchange time** — from catcher receipt to release of the throw (`ball_eventcode` 3).
- **Flight time** — from release of the throw to infielder receipt.

Throw velocity. Ball speed during the flight window, computed from successive `ball-positions` records as the three-dimensional displacement between consecutive frames divided by the elapsed time, then converted to miles per hour. Because tracking noise inflates single-frame maxima, three summaries are retained: the maximum, the 95th percentile, and the mean across the flight window. The 95th percentile is used in modeling.

Contested attempt. An indicator for whether the defense made a throw to second base on the attempt. This is recorded as a covariate rather than used as a filter (see Data Filtering).

Data filtering rules. Candidate steal attempts were first identified using the base-state information from the lineups table. Plays were restricted to situations with a runner on first base and second base unoccupied, so the advance could be attributed to the runner on first. Plays with batted balls, pickoff throws, walks, or hit-by-pitches were removed because the runner's advance may have been caused by another game event rather than an independent steal attempt.

After this initial filtering, tracking data was used to confirm that the runner actually advanced toward second base. A play was kept if the runner made substantial progress toward second base, using displacement toward second and runner closing speed relative to pitch release. A catcher throw was not required for a play to qualify as an attempt because that would remove uncontested steals and bias the sample toward only contested plays.

Modeling details. The modeling dataset included only confirmed steal attempts with a clean safe/out outcome and complete values for the four model inputs: runner top speed, runner jump, catcher pop time, and catcher throw velocity. Safe plays were coded as 1, and caught stealing plays were coded as 0. The data was split into a 70% training set and a 30% test set using a fixed random seed for reproducibility. Three models were compared: logistic regression, a decision

tree, and a random forest. Each model produced a predicted probability that the runner would be safe. Classification thresholds were selected on the training set by testing cutoffs from 0.05 to 0.95 and choosing the threshold with the highest balanced accuracy.

The models were evaluated on the held-out test set using accuracy, sensitivity, specificity, balanced accuracy, and AUC. Logistic regression was selected as the final Safe% model because it produced smoother and more interpretable probability estimates for a scoreboard setting. After model selection, the final logistic regression model was fit to the full clean modeling dataset to generate Safe% values for scoreboard examples.

Supplemental leaderboards not shown in paper:

BEST RUNNER JUMPS			
Most speed gained in the first 0.4 seconds			
RANK	PLAYER	GAME	VALUE
#1	Player VAS-1487	D199: JGO-VAS	8.09 mph gained
#2	Player ARN-0727	D177: AVV-ARN	7.83 mph gained
#3	Player RRM-1635	D187: RRM-VAS	7.41 mph gained
#4	Player VNG-4196	D149: VNG-VAS	7.35 mph gained
#5	Player VAS-1487	D202: PHD-VAS	7.34 mph gained
#6	Player ANI-1431	D209: QHX-ANI	7.06 mph gained
#7	Player VAS-1487	D198: JGO-VAS	6.92 mph gained
#8	Player LRQ-9850	D181: LRQ-ARN	6.8 mph gained
#9	Player ANI-2830	D128: MEX-ANI	6.62 mph gained
#10	Player VAS-7896	D185: RRM-VAS	6.58 mph gained

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HARDEST CATCHER THROWS			
Highest catcher throw velocities on steal attempts			
RANK	PLAYER	GAME	VALUE
#1	FNQ-7853	D163: FNQ-PHD	81.98 mph
#2	VAS-6967	D209: PHD-VAS	80.97 mph
#3	SZJ-5082	D166: SZJ-ANI	80.76 mph
#4	VHA-5456	D148: VHA-PHD	80.16 mph
#5	AVV-5369	D178: AVV-ARN	80.1 mph
#6	VAS-3698	D177: VAS-PHD	79.69 mph
#7	ARN-0174	D133: ARN-VAS	79.67 mph
#8	RYU-6946	D083: RYU-PHD	79.43 mph
#9	ANI-7970	D127: MEX-ANI	79.2 mph
#10	PHD-5619	D138: HYG-PHD	78.91 mph
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Supplemental model graphic not shown in paper:

GOOD TAG!	
This play beat expectation!	
SAFE% 75%	
Actual Result: Out Runner: PHD-5994	Model Prediction: Safe Catcher: XPO-7814
Top Speed: 18.5 mph Jump: 4.3 ft/s gained	Pop Time: 1.95 sec Throw Velo: 72 mph
STOLEN BASE SPOTLIGHT MODEL EXPECTATION BOARD	